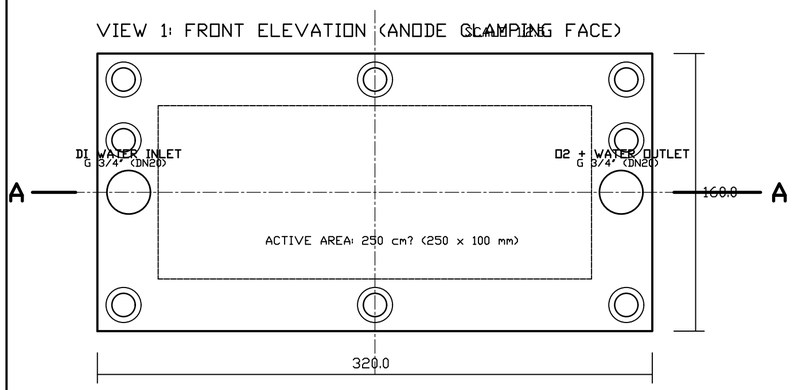
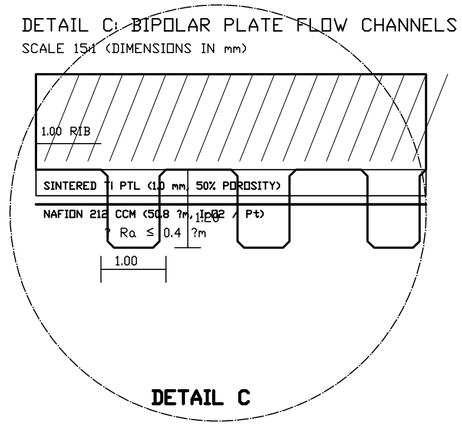
VIEW 1: FRONT ELEVATION (ANODE CLAMPING FACE)

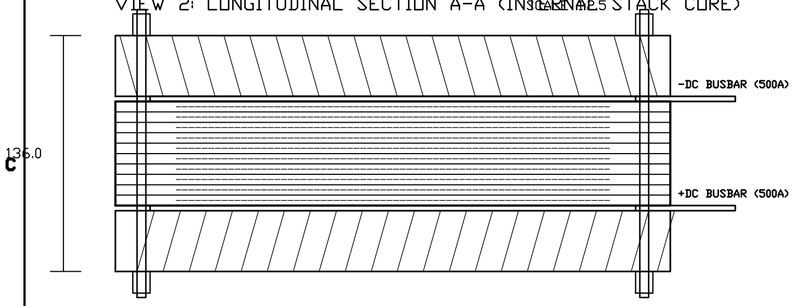


DETAIL C: BIPOLAR PLATE FLOW CHANNELS & MEMBRANE INTERFACE

SCALE 15:1 (DIMENSIONS IN mm)



VIEW 2: LONGITUDINAL SECTION A-A (INTERNAL STACK CORE)



ENGINEERING BILL OF MATERIALS (BOM)

ITM	COMPONENT DESCRIPTION	MATERIAL SPECIFICATION	QTY	STANDARD / CODE
01	Anode Heavy End Clamping Plate	AISI 316L Stainless Steel (35mm)	01	ASTM A240
02	Cathode Heavy Clamping Plate	AISI 316L Stainless Steel (35mm)	01	ASTM A240
03	Anode Current Collector Plate	C11000 ETP Copper, 5?m Au Plated	01	ASTM B152
04	Cathode Current Collector Plate	C11000 ETP Copper, 5?m Au Plated	01	ASTM B152
05	Bipolar Plates (Anode/Cathode)	Titanium Grade 2 (4.5mm Machined)	11	ASTM B265
06	Porous Transport Layer (PTL)	Sintered Porous Titanium (1.0mm)	10	50% Porosity
07	Gas Diffusion Layer (GDL)	Toray Carbon Paper TGP-H-060	10	PTFE 5wt%
08	Catalyst Coated Membrane (CCM)	Nafion 212 (50.8?m) IrO2/Pt-C	10	1.5/0.4 mg/cm²
09	Perimeter Elastomer Gasket	Fluorosilicone / EPDM (0.8mm)	22	Shore A 65
10	High-Tensile Clamping Tie-Rods	Grade 8.8 High-Tensile Steel M12	08	DIN 976 / ISO 898
11	Belleville Disc Spring Packs	50CrV4 Heavy Duty Disc Springs	16	DIN 2093 Group 2
12	Hexagonal Clamping Nuts	AISI 316 Stainless Steel M12x1.75	16	DIN 934 / ISO 4032
13	Fluid Header Port Adapters	316L SS G 3/4" Male BSPP to Swagelok	02	ISO 228-1
14	Manifold O-Ring Face Seals	FKM Fluoroelastomer (Viton 70)	04	AS568-115

OPERATING DESIGN PARAMETERS & GENERAL ENGINEERING NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETERS (mm) UNLESS OTHERWISE SPECIFIED. THIRD-ANGLE PROJECTION.

2. ACTIVE ELECTROCHEMICAL AREA: 250 cm² (250 mm x 100 mm). RATED STACK HYDROGEN OUTPUT: 1.66 kg H2/hr (18.6 Nm³/hr @ 30 bar).

3. OPERATING REGIME: CURRENT DENSITY 0.1 – 2.5 A/cm² (500 A DC @ 2.0 A/cm²); NOMINAL STACK VOLTAGE: 18.8 V DC @ 80 °C, 30 bar.

4. CLAMPING PRELOAD: TIE-RODS TORQUED TO 15.0 kN PER BOLT IN 3 EQUAL CROSS-PATTERN STAGES (5 -> 10 -> 15 N?m) TO ACHIEVE 1.5 MPa UNIFORM CONTACT PRESSURE.

5. HELIUM LEAK INTEGRITY: MAX LEAKAGE RATE <= 1.0E-6 mbar?L/s AT 35 bar DIFFERENTIAL PROOF PRESSURE BEFORE ELECTROCHEMICAL ACTIVATION.

6. BIOFOULING SENSING: MONITORED VIA IN-SILICO DIELECTRIC EQUIVALENT CIRCUIT. PROACTIVE 35% LOAD CURTAILMENT TRIGGERED WHEN ?V > 30 mV RESIDUAL.

7. FLOW CHANNELS (50 PARALLEL): MACHINED VIA HIGH-PRECISION CNC MILLING WITH SURFACE FINISH Ra <= 0.4 ?m TO PREVENT LOCALIZED TWO-PHASE OCCLUSION.

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JADAVPUR UNIVERSITY & SISTER NIVEDITA UNIVERSITY

DEPT. OF CHEMICAL ENGINEERING | DEPT. OF MICROBIOLOGY & BIOTECHNOLOGY

10-CELL INDUSTRIAL PEM ELECTROLYZER STACK GA & DETAILS

PROJECT: 1:1 INDUSTRIAL PEM GREEN HYDROGEN DIGITAL TWIN (ALTFS-2026)

DESIGNED BY: SOURISH SENAPATI (CHEM ENG, JU) MICROBIOLOGY: SRESTHA BISWASH (MICROBIO, SNU)	REV: 01	SHEET: 1 OF 1
DWG NO: DWG-PEM-H2-2026-001	SCALE: AS NOTED	DATE: 2026-11-05